

THOMAS REESE

Software Engineer · Full-Stack & AI Systems

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SUMMARY

Full-stack engineer (4+ yrs) shipping AI-native systems on AWS Bedrock and the Model Context Protocol. Currently at Envoy, where I architected Codebase Intelligence — an 87,500-chunk vector index serving 22 repos at \$0.03/month. Production stack spans Ruby on Rails, Ember.js, React 19, and Python. Comfortable from infra (Pulumi, Terraform) to model eval (RAG, MCP tool use, embeddings).

EXPERIENCE

Envoy | Software Engineer, Visitor Management

San Francisco, CA · Nov 2024 – Present

- Architected **Codebase Intelligence**, a RAG platform indexing 22 repos (~87,500 Tree-sitter AST chunks) on **Amazon S3 Vectors** and **Bedrock Knowledge Bases** (Titan Text Embeddings V2), with a Python **MCP server** exposing six retrieval tools to Anthropic **Claude**. Hit **0.785 top-1 recall** at **\$0.03/month**, retiring 6–8 GB per-engineer local vector DBs.
- Migrated the visitor-management dashboard from **Ember/Glimmer to React 19** across six surfaces (profile, printers, screeners, visits, matching-rules, documents) using **Module Federation 2.0**, **TanStack Query v5**, and **Radix UI**, shipping incrementally behind feature flags.
- Shipped **Visitor Profile Merge** and **Block List V2** end-to-end across two quarters: Rails APIs, React merge UI, and the identity-matching lifecycle (expiration, disable toggle, search, deletion), eliminating manual support handling for duplicate-visitor cleanup.
- Replaced fragile SKU-name string matching with a **ChargeBee feature-code subscription gate** on the Visitors Basic plan: counter table, gate-state API, 75% / 100% threshold mailer, and reset cron, spanning Rails and Ember in one sprint.
- Authored **AgentLock**, a phase-based permission model governing the 14 AI tools embedded in the Visitors product, and scoped the Visitors AI roadmap (drafting, fuzzy matching, NL analytics, audit log, smart dedup).
- Resolved customer-facing bugs blocking Fortune 500 accounts (**Blue Origin**, **Tesla**, **NVIDIA**) using TDD: **RSpec** on Rails, **Vitest** + **MSW** on React, with staging verification via Kubernetes pod exec.
- Authored the org's **AI-tooling onboarding guide**, a **six-layer agent memory architecture** reference, and a coding-agent observability writeup, adopted across teams as Envoy's internal AI playbook.
- Shipped at **5+ PRs/week** across nine repos on a stacked-PR (**Graphite**) and worktree-based workflow, covering Rails, Ember (Octane / Glimmer / GTS), React 19, **Cerbos** authorization, and Pulumi / Terraform IAM, with zero production rollbacks over 18 months.

Sprout Social | Front-End Engineer

Chicago, IL · May 2022 – Apr 2024

- Migrated core front-end components from **Flow to TypeScript**, eliminating a class of runtime type errors and unblocking strict-mode adoption across the codebase.
- Built and maintained the **Seeds** shared React component library, standardizing UI primitives across product teams and cutting per-feature build time by ~3 days.
- Built a React UI for **bulk sentiment reclassification** (1,000 messages per batch) using **Redux** and virtualized lists, cutting analyst handle time ~30% per batch.
- Built a **Query Builder** in React + TypeScript backed by **OpenAI GPT** keyword suggestions with debounced search, lowering query-setup time for social-listening analysts.
- Shipped a **faceted message-exploration tool** (React Context, Redux, React Router) with virtualized lists, lifting session engagement ~30% on the listening dashboard.

The Walt Disney Company | Software Engineering Intern

Remote · May 2021 – Nov 2021

- Shipped React and responsive-CSS fixes across multiple **ABC News** station sites: navigation restructure, broken-link remediation, and mobile UI/UX polish.
- Built **ESPN's Banner-Maker**: reusable React components with Redux state and team / player API integration, cutting editorial banner-creation time ~35%.
- Owned the React front-end for a **Student Loan Refinance** product: responsive UI, client-side validation, and back-end API integration, cutting form-validation errors ~60%.
- Drove **WCAG** accessibility work with **Applitools**: screen-reader audits and semantic-HTML refactors that raised automated a11y coverage ~70%.

Credit Karma | Full-Stack Web Developer Intern

San Francisco, CA · Jan 2020 – Dec 2020

- Built internal full-stack tooling across **React**, **Node.js**, **Ruby**, and **Python** services on **AWS**, focused on infrastructure automation and front-end developer experience.
- Partnered with platform engineers on shared infrastructure improvements, contributing changes to internal libraries and CI pipelines.

SKILLS

Languages	TypeScript, JavaScript (ES6+), Python, Ruby, HTML5, CSS3, SQL, Flow (legacy)
Frameworks & UI	React 19, Next.js, Ember.js (Octane / Glimmer / GTS), Ruby on Rails, FastAPI, Node.js, Express, Tailwind, Radix UI, SCSS, Luxon, Three.js, Framer Motion
AI & Data	Model Context Protocol (MCP), Anthropic Claude SDK, OpenAI SDK, Gemini Flash, Amazon Bedrock, S3 Vectors, Titan Text Embeddings V2, Tree-sitter AST chunking, ChromaDB, Pinecone, LangChain, Optuna
Cloud & Infra	AWS (Bedrock, S3, IAM), Pulumi, Terraform, Kubernetes, Docker, OpenTelemetry, Datadog, Vercel
Data Stores	PostgreSQL, Redis, Kafka, Sidekiq, MongoDB
Build & State	Module Federation 2.0, Vite, Webpack 5, TanStack Query v5, Redux Toolkit, Axios, MSW v2, GraphQL, REST
Testing	Vitest, RSpec, Playwright, Jest, Cypress, MSW v2
Workflow & Sec	Graphite (stacked PRs), GitHub Actions, CircleCI, Cerbos, OAuth, JWT, WCAG accessibility

EDUCATION

Dominican University
B.S. Computer Science

River Forest, IL
Aug 2018 – May 2022

PROJECTS

TraderDan — Automated Trading System *Python · Next.js · FastAPI · Anthropic Claude · TradeLocker · PostgreSQL · Optuna · ChromaDB*
[\[github\]](#)

Production trading system managing real capital across five active prop-firm accounts, trading XAUUSD on H1/M15 via TradeLocker. Architecture spans a backtest engine, live signal generator, FastAPI webhook execution server with prop-firm risk gates (daily-loss, drawdown, ATR-spike, news blackout, weekend closer), a 14-agent AI team led by an OpenClaw CEO orchestrator (Anthropic Claude tool use, Gemini Flash routing), an eval harness with drift detection, semantic memory in ChromaDB, and a Next.js 15 dashboard with 17 pages for live monitoring. 5,696+ tests on a stacked-PR workflow.

Blue Ridge Canine Academy [\[link\]](#) *React · Node.js · Express · MongoDB · Acuity Scheduling · Figma*
Freelance build (\$800) for a local dog trainer: a full-stack scheduling and payments platform. React frontend, Node.js / Express backend, MongoDB persistence. Wireframed in Figma and hosted on GoDaddy.

HONORS & AWARDS

- Won **1st place**, Google × Code Nation Hackathon.

- Won the **24-hour Code for Good Hackathon** by JP Morgan, leading a front-end team that shipped a dynamic site for the Girls in the Game non-profit.
- Won **2nd place** at the ACCA Programming Contest (Associated Colleges of the Chicago Area), the highest Dominican University finish to date.
- Completed **MLT Career Prep** (Management Leadership for Tomorrow).
- Mentored high school students in STEM and technology at **Code Nation**.
- Featured in a **Credit Karma** commercial on personal credit recovery; also appeared in a **JumpStart** rebranding spot on early-career recruiting.